

Aarush Mane

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EDUCATION

Princeton University

B.S.E. in Computer Science

Princeton, NJ

Expected 05/2030

– Dual Enrollment Coursework: Single Variable Analysis (MAT215), Linear Algebra w/ Intro to Proofs (MAT217).

Robbinsville High School

Robbinsville, NJ

High School Diploma | Weighted GPA: 4.682/5.0, Unweighted GPA: 3.977/4.0 | SAT: 1530

09/2022 – 06/2026

– Honors: National Merit Scholarship Winner, AP Scholar with Distinction, NJ State Seal of Biliteracy (Spanish).

EXPERIENCE

FIRST Robotics Competition Team 2590, Nemesis

Robbinsville, NJ

Software Development & Strategy Lead

10/2023 – 06/2026

- Led technical strategy and statistical performance modeling across **65** members, building **Excel/Sheets** analytics pipelines to pitch real-time match play and secure **5th** place out of **600+** teams at the World Championship.
- Architected computer vision algorithms in **Java** and **OpenCV**, implementing Oculus Quest coprocessor localization for **3D** pose tracking and AprilTag PnP pose estimation, pitching design to corporate judges to earn the Innovation in Control Award.
- Built autonomous trajectory wrappers in **Java** to parse **JSON** motion profiles, synchronizing subsystem command scheduling to eliminate kinematic lag during high-speed field maneuvers.
- Developed "Argus", an autonomous Slack AI agent (**MiniMax LLM**, **Nanobot** framework); integrated **Python** API scrapers and custom **GitHub** workflows to automatically stream scouting data and match logs into analytical sheets and graphics.
- Mentored **10+** junior developers in **Java** and autonomous control architecture, enforcing **Git** branching workflows, unit testing pipelines, and strict pull request code review standards.

FIRST Tech Challenge Team 22348, REAPR

Robbinsville, NJ

Founder & Software Mentor

10/2022 – 06/2026

- Secured **\$16,000+** in corporate and local business sponsorships through targeted partnership proposals, establishing sustainable funding pipelines for a **15-member** community robotics initiative.
- Managed ongoing relationships with institutional partners and school district administrators, increasing regional STEM outreach and community participation by **30%**.
- Instructed student members on training custom computer vision and object detection models using **TensorFlow** along with autonomous routines in **Java**, while building real-time data pipelines on **AppSheet** to guide tactical decisions.

NJ Governor's School of Engineering & Technology (NJ GSET)

Rutgers University, NJ

Research Scholar

07/2025 – 07/2025

- Collaborated on a **5-person** research team mentored by a graduate researcher to model autonomous path-tracking algorithms.
- Pitched technical research at the GSET Final Symposium before an audience of **300+** university faculty, corporate sponsors, and industry partners.

iStart Valley

Remote

Financial Projections & Business Strategy Intern

01/2025 – 06/2025

- Pitched venture decks to panels of industry judges and investors for "BioPack" (**\$500k valuation**), securing placement as a Global Semifinalist in the iStart Valley Startup Pitch Contest.
- Engineered comprehensive financial projections, unit economics, and **TAM/SAM/SOM** market analyses to articulate clear value propositions to corporate stakeholders and early-stage partners.
- Recognized with the Gold Presidential Volunteer Service Award for completing **265 hours** of executive-level technical writing, stakeholder management, and community outreach.

PUBLICATIONS

IEEE Xplore | IEEE MIT Undergraduate Research Technology Conference (URTC)

Presented Oct. 2025

Co-Author | PID Control Optimization and Trajectory Tracking for Autonomous Differential Drive Robots

[IEEE Link](#)

- Engineered an enhanced, non-linear PID controller featuring moving integral windows, velocity-dependent target yaws, and anti-windup saturation limits in **Python** for low-cost autonomous differential drive rovers.
- Simulated path generation algorithms (A* vs. Dijkstra) and **ROS GazeboSim LiDAR** obstacle avoidance to model real-world resistive forces, variable system dynamics, and node-based room traversal.
- Validated controller efficacy at the Emil Buehler Aerospace Lab using **TurtleBot Vicon motion capture tracking** and statistical matched-pairs t-tests ($p < 0.01$) to demonstrate low-cost hardware viability.

TECHNICAL SKILLS

Partnerships & Strategy : Sponsorship Pitching, Corporate Outreach, Technical Pitching, TAM/SAM/SOM

Languages & Data : Java, Python, MATLAB, LaTeX, JSON, HTML/CSS, Excel, Google Sheets, Hypothesis Testing

Tools, Frameworks & Hardware : OpenCV, TensorFlow, ROS, GazeboSim, Git, GitHub Actions, Vicon, AppSheet

Spoken Languages : English (Native), Spanish (Fluent), Telugu (Fluent)